Cantera vs NN | Test IC 18 | NRMSE (util.py)

T0 = 1069.1 K P0 = 2.170 atm phi0 = 1.544 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	0.0255	0.115	0.793	0.959	0.336	1.94	0.0671	1.02	1.19	0	Complete
2 P-only DeepONet10	**10k**	0.067	0.161	0.806	1.05	0.768	0.82	0.0892	1.54	2.36	0	Complete
2 P-only DeepONet10	**long**	7.36e-03	0.0402	0.184	0.374	0.17	0.0525	0.018	0.468	0.667	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.029	0.0992	0.314	0.756	0.462	0.297	0.051	1.31	1.86	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.0118	0.0562	0.249	0.529	0.241	0.0846	0.0261	0.698	0.953	0	Complete
4 T/P DeepONet10	**10k**	0.0652	0.169	1.06	1.06	0.778	1.03	0.0828	1.98	5.67	0	Complete
4 T/P DeepONet10	**long**	—	—	—	—	—	—	—	—	—	—	**Stop 0.552 ms**
5 ANN10 + T/P DeepONet10	**10k**	0.0681	0.149	1.03	1.02	0.693	1.02	0.0796	1.62	2.2	0	Complete
5 ANN10 + T/P DeepONet10	**long**	0.371	0.261	7.26	0.43	0.129	5.42	0.155	0.327	0.576	0	Complete
6 T/P/ϕ DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 0.250 ms**
6 T/P/ϕ DeepONet10	**long**	0.464	0.331	8.77	2.73	0.568	5.54	0.235	27.6	2.16e+04	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	0.304	0.175	3.44	0.629	0.833	8.52	0.265	2.04	5.77	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.389	0.223	7.2	1.26	0.0564	5.87	0.181	0.25	0.299	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.